

FINAL PROJECT REPORT

Online Learning Engagement and Student Performance Analysis

1. Introduction

1.1. Project overview

Online Learning Engagement and Student Performance Analysis is a Power BI data analytics project that analyses student-level online education data to understand learning engagement, academic performance, and dropout behaviour. The project uses student information related to learning activities, academic results, educational background, and other available characteristics.

Power BI is used to clean, transform, analyse, and visualize the data through interactive dashboards and reports. The analysis focuses on important factors such as engagement level, total clicks, average score, performance level, final result, risk level, dropout patterns, gender, region, and highest education level, based on the fields available in the dataset.

The main purpose of this project is to identify useful patterns and trends in student learning behaviour and performance. The interactive Power BI dashboard helps users understand the data easily and supports data-driven analysis of student engagement and academic outcomes.

1.2. Objectives

- To assess overall student engagement and academic performance.
- To identify factors influencing student success, failure, and dropout.
- To analyze dropout patterns and identify students at potential risk.
- To compare engagement and performance across different student groups.
- To examine the relationship between learning activity and academic performance.
- To analyze performance differences based on region and educational background.
- To monitor student pass and dropout rates.
- To support data-driven strategies for student retention and intervention.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

I am: An educational institution or online learning platform.

I am trying to: Analyse students' online learning engagement, academic performance, and dropout behaviour using student-level data.

But: It is difficult to identify the key factors associated with student performance, engagement, and dropout patterns.

Because: The available data contains information such as engagement level, total clicks, assessment performance, final results, educational background, region, and risk level.

Which makes me feel: It is challenging to identify students or groups requiring additional support and make timely decisions to improve academic performance and reduce dropout rates.

2.2. Project Proposal (Proposed Solution)

The proposed solution is an interactive **Power BI dashboard and report** designed to analyse online learning engagement and student performance. It brings student-level data into a single visual environment, allowing performance, engagement, risk, dropout, and demographic patterns to be analysed using charts, cards, and filters.

Project Scope includes:

- Student engagement and learning activity analysis.
- Academic performance and final result analysis.
- Risk and dropout analysis.
- Comparison based on gender, region, and education level.
- Analysis of total clicks and engagement levels.
- Monitoring of pass and dropout rates.

System Requirements

Basic System Requirements:

Operating System: Windows 10 or Windows 11 (64-bit)

Processor: Intel Core i3 / AMD Ryzen 3 or higher

RAM: Minimum 8 GB

Storage: At least 5 GB of free space

Display: 1366 × 768 resolution or higher

Database: MySQL / SQL Server or any compatible SQL database

SQL Tool: MySQL Workbench / SQL Server Management Studio

Power BI: Power BI Desktop (latest version recommended)

2.3. Initial Project Planning

The project is completed individually. The following product backlog summarizes the planned work.

Sprint	Functional Requirement	User Story Number	User Story / Task Story	Points	Priority
Sprint-1	Data Understanding	USN-1	As a project member, I can understand the online education dataset and identify important student fields.	2	High
Sprint-1	Data Cleaning	USN-2	As a project member, I can clean and prepare the student data for analysis in Power BI.	2	High
Sprint-1	Data Preparation	USN-3	As a project member, I can prepare the required data fields and measures for the dashboard.	2	High
Sprint-2	Student Performance	USN-4	As a user, I can view student performance using average score and performance level.	2	High
Sprint-2	Student Engagement	USN-5	As a user, I can analyze student engagement using engagement level and average clicks.	2	High
Sprint-2	Final Results	USN-6	As a user, I can view Pass, Fail, Withdrawn, and Distinction results.	2	Medium
Sprint-3	Risk Analysis	USN-7	As a user, I can analyze student risk levels using the available risk categories.	2	High
Sprint-3	Dropout Analysis	USN-8	As a user, I can compare dropout rates	2	High

			across educational backgrounds.		
Sprint-3	Demographic Analysis	USN-9	As a user, I can analyze student information by gender, region, and highest education.	1	Medium
Sprint-3	Dashboard Development	USN-10	As a user, I can explore student performance, engagement, risk, and dropout information in Power BI.	2	High

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

The main raw data source for the project is the file named online_education_dataset.csv. The dataset is used as the student-level source for the Power BI analysis. The project uses the available student information to study engagement, performance, final results, risk, dropout, demographic characteristics, and educational background.

- **Dataset Name:** online_education_dataset.csv
- **Dataset Format:** CSV
- **Dataset Size:** 32953

3.2. Data Quality Report

The project uses student-level online education learning data. The dataset contains information required to study student engagement, academic performance, final results, risk, dropout behaviour, educational background, region, and other student characteristics.

Important fields used in the analysis include engagement level, total clicks, average score, performance level, final result, risk level, gender, region, and highest education.

3.3. Data Exploration and Preprocessing

The dataset was explored to understand student engagement, academic performance, risk levels, dropout patterns, and demographic characteristics.

The relevant data was prepared in Power BI using appropriate measures, categories, filters, and visualizations.

Key fields such as engagement level, average clicks, average score, final result, risk level, gender, region, and education level were analysed.

The analysis helps identify patterns in student behaviour and performance and supports better understanding of dropout and engagement trends.

4. Data Visualization

4.1. Framing Business Questions

No.	Business Question	Visualization / Analysis
1	What is the distribution of final student results?	Donut chart showing Pass, Fail, Withdrawn, and Distinction results.
2	Which regions have the highest number of student records?	Clustered bar chart showing total students by region
3	How does average score vary by highest education level?	Clustered column chart showing average score by education level
4	What is the relationship between engagement level and average total clicks?	Column chart showing average total clicks by engagement level.
5	Which risk level has the highest dropout rate?	Bar chart showing dropout rate by risk level.
6	How does performance level differ by gender?	Stacked column chart showing performance-level distribution by gender.
7	How are final results distributed across engagement levels?	Decomposition table showing final result by engagement level
8	What is the overall average score?	KPI CARD

4.2. Developing Visualizations

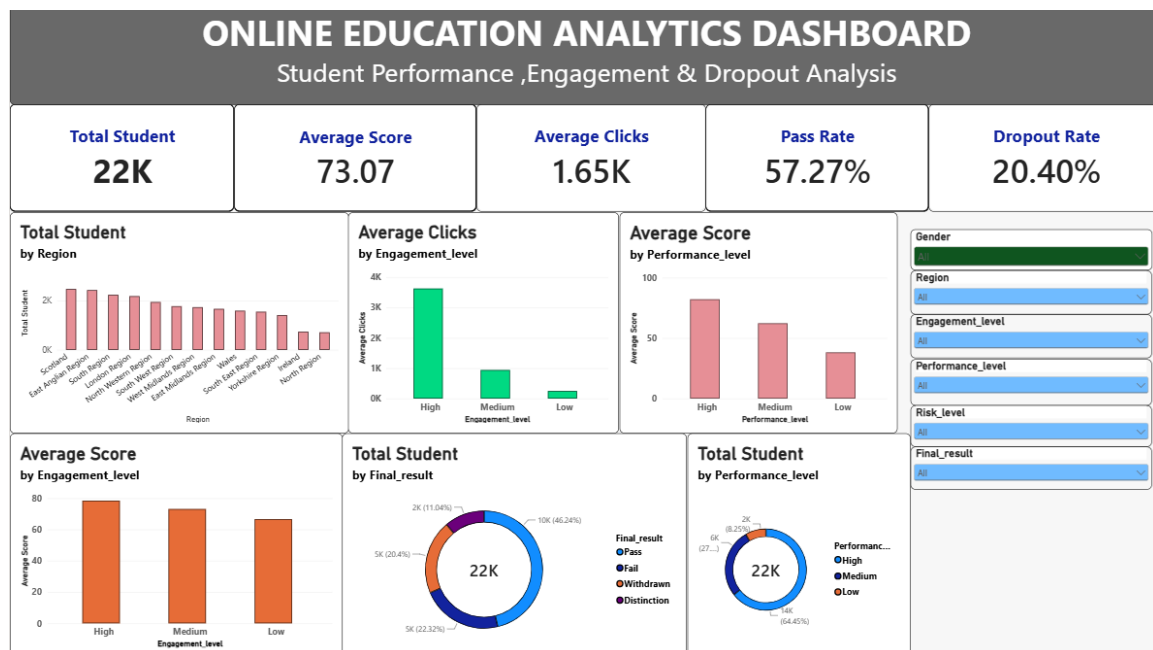
The Power BI report uses interactive visualizations to present student data clearly and support easy analysis. The dashboard includes key measures such as student count, average score, engagement level, average clicks, final results, risk categories, pass rate, and dropout rate.

- Student count and summary information.
- Average score by region.
- Average clicks by engagement level.
- Final result distribution.
- Risk category analysis.
- Dropout and pass rates by highest education.
- Gender- and region-based analysis.

5. Dashboard

5.1. Dashboard Design

The Power BI dashboard provides an interactive overview of student performance, engagement, and dropout patterns. The dashboard shows **22K total students**, with an **average score of 73.07**, **average clicks of 1.65K**, a **pass rate of 57.27%**, and a **dropout rate of 20.40%**. It compares student distribution across regions and shows that students with **High engagement** have the highest average clicks. The dashboard also compares average scores across engagement and performance levels and presents the distribution of students by final result and performance level. Interactive filters for **gender, region, engagement level, performance level, risk level, and final result** allow users to explore the data in greater detail.

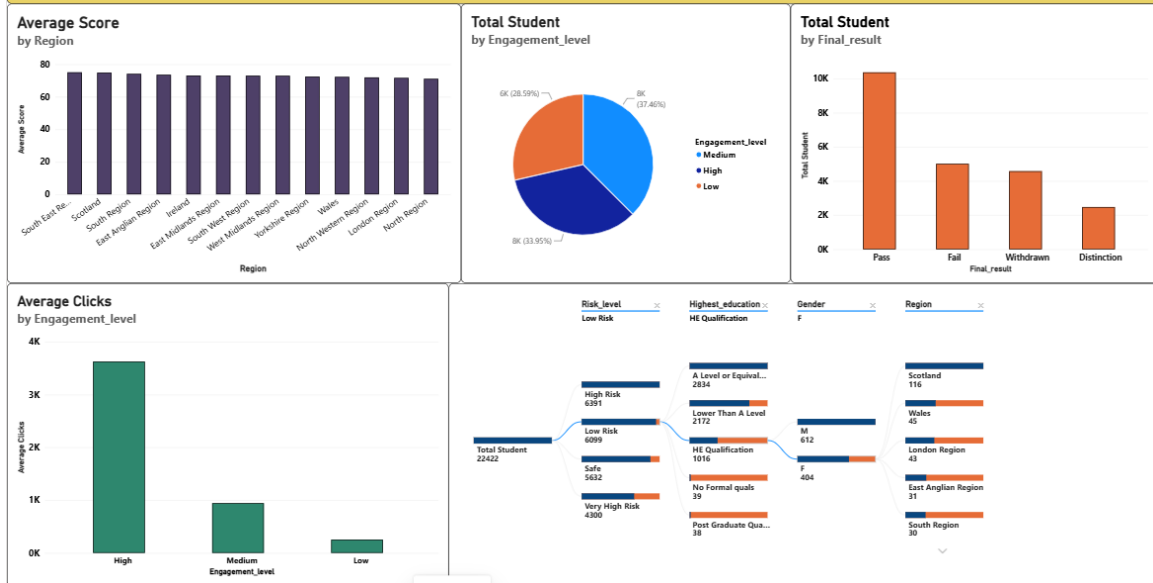


6. Report

6.1-Global online learning-Student Performance &Engagement Overview

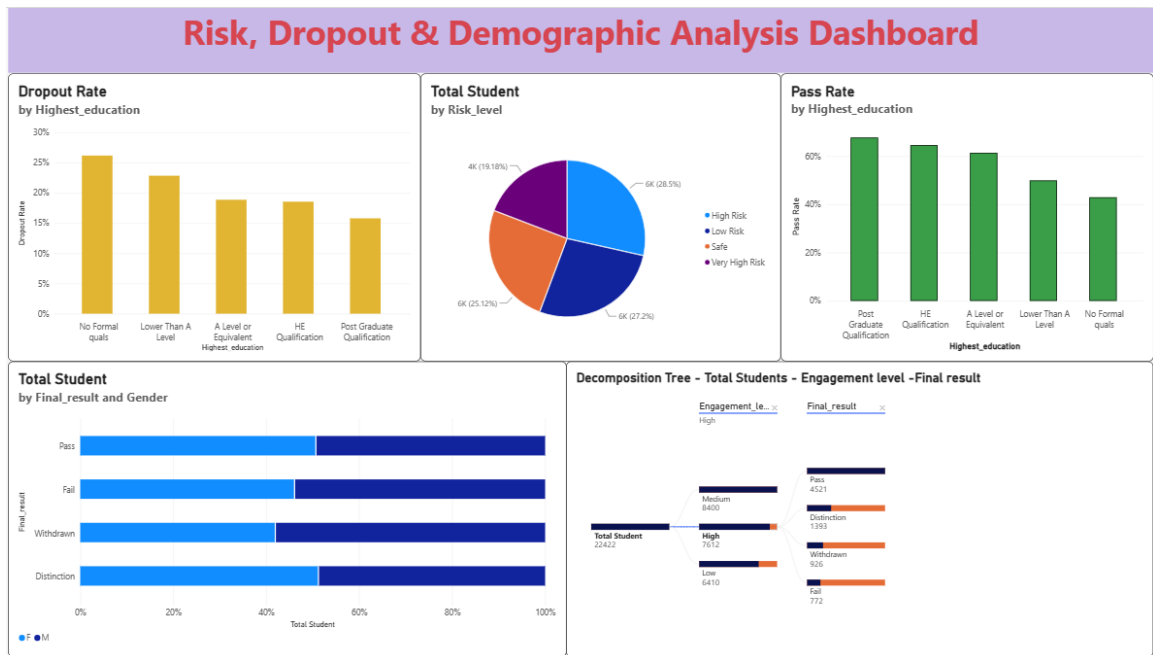
The **Global Online Learning – Student Performance & Engagement Overview** dashboard provides a detailed view of student engagement, academic performance, and risk levels. The dashboard shows average scores across regions, student distribution by engagement level, and final results such as Pass, Fail, Withdrawn, and Distinction. It also compares average clicks across different engagement levels, showing higher activity among highly engaged students. The interactive analysis further provides information on risk level, highest education, gender, and **region**, allowing student groups to be examined based on different characteristics. Overall, the dashboard helps identify patterns in student engagement and performance and provides a clear view of factors related to student outcomes.

Global Online Learning- Student Performance & Engagement Overview



6.2- Risk, Dropout & Demographic Analysis Dashboard

The Risk, Dropout & Demographic Analysis Dashboard provides insights into student dropout, pass rates, risk levels, and demographic characteristics. The dashboard compares dropout and pass rates across different education levels and shows the distribution of students by risk category. It also presents the relationship between final results and gender, along with a decomposition of student engagement levels and final results. These visualizations help identify patterns in student risk, academic outcomes, and dropout behaviour.



7. Performance Testing

7.1. Utilization of Data Filters

Filter / Slicer	Purpose
Gender	Compare outcomes by gender
Region	Focus the analysis on specific regions
Engagement Level	Filter High, Medium, or Low engagement students
Performance Level	Filter High, Medium, or Low performance students
Risk Level	Filter Low Risk, Safe, High Risk, or Very High Risk

7.2. Number of Calculation Columns / Measures

Five core report measures are identifiable in the supplied PBIX report definition: Total Students, Average Score, Average Clicks, Dropout Rate, and Pass Rate. The preprocessing documentation additionally states that a unique index column was added to identify each record. A separate calculation-column count is not explicitly exposed in the supplied report documentation, so no additional number is asserted here.

7.3. Number of Visualizations

Visual Element	Count
Card visuals / KPI cards	10
Slicers	6
Column charts	3
Donut chart	1
Pie chart	1
Text boxes	3
Total visuals across supplied PBIX definition	24

8. Conclusion/Observation

The project demonstrates how Power BI can be used to analyse online learning engagement and student performance using student-level data. The dashboard combines key factors such as engagement, average clicks, average score, final results, risk level, dropout rate, pass rate, region, gender, and education level.

The visualizations make it easier to compare student groups and identify patterns in engagement, academic performance, risk, and dropout behaviour. The analysis provides useful insights that can help educational institutions understand student outcomes and support data-driven decisions for improving student performance and reducing dropout.

9. Future Scope

- Include additional student activity and learning behaviour data.
- Add time-based analysis of student engagement and performance.
- Implement predictive analysis to identify potential dropout risks.
- Provide detailed drill-down analysis for different student groups.
- Connect the dashboard to regularly updated data sources.
- Add new Power BI measures and interactive visualizations.
- Enhance reporting features for effective student retention monitoring.

10. Appendix

GitHub & Project Demo Link

GitHub Link: <https://github.com/sandhyach0002/Online-Learning-Engagement-and-Student-Performance-Analysis.git>

Project Demo Link: <https://github.com/sandhyach0002/Online-Learning-Engagement-and-Student-Performance-Analysis/blob/main/Project%20Documentation%20and%20Demonstration/demo%20video1.mp4>

